Sahaj Bhadja

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Summary

Software engineer with 2+ years building **Python/Java backend systems**, **low-latency identity APIs**, and **distributed decisioning services** with strong ownership of reliability, observability, and production quality.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 - Present*
- Built **high-throughput backend services** for 500+ users, reducing query latency from **1.2s to 350ms** through SQL optimization and API path tuning.
- Implemented **metrics-driven validation workflows** for 5K+ monthly jobs, improving production reliability and reducing escalation risk across critical data flows.

Grad8 **LA, California**
Software Engineer Intern *June 2025 - August 2025*
- Developed **full-stack features and service endpoints** handling 10K+ daily requests, translating product requirements into robust backend APIs and UI workflows.
- Improved **release quality and on-call readiness** with automated CI/CD checks and structured defect triage, cutting turnaround by **60%**.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 - May 2025*
- Built **data processing pipelines** over 100K+ trajectory records for reproducible experiments, enabling accurate model evaluation and reliable decision analytics.
- Published **technical design and behavior docs** pre-release, improving cross-functional alignment and reducing ambiguity during integration and review cycles.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 - July 2024*
- Engineered **distributed backend workflows** using AWS Glue and S3, improving response times by **30%** while supporting resilient, high-volume data pipelines.
- Maintained **20+ automated test suites** with Jenkins, PyTest, and Mockito, increasing pipeline efficiency by **68%** and reducing regression defects.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 - August 2023*
- Developed **MERN + WebSocket modules** for 2M+ synchronized records across 8 workflows, improving consistency of user sessions and event-driven updates.
- Automated **Docker-based deployments** with Jenkins, decreasing release time by **50%** and improving stability across shared environments.

Skills

Languages: **Python**, **Java**, SQL, JavaScript/TypeScript, C++, Bash
Backend & APIs: REST services, Spring Boot/Flask patterns, distributed service integration, asynchronous processing, auth/verification flows
Data & Cloud: MySQL/SQL Server, query optimization, AWS (S3, Glue), containerized deployment, Kubernetes fundamentals
Reliability: CI/CD pipelines, test automation, metrics/monitoring, incident debugging, peer code reviews
Collaboration: Agile development, cross-functional communication, requirement translation, technical documentation

Projects

Identity Risk Decisioning API | *Python, FastAPI, MySQL, AWS*
- Built a **low-latency decisioning service** for user verification and risk scoring with cached rule evaluation, improving p95 response times and service availability.

Chat2SQL Code | *Java, TypeScript, SQL, Docker*
- Developed a **schema-aware query assistant** across 10+ datasets with validation and explainable outputs, reducing manual debugging and analyst turnaround.

Linguistix Code | *Python, AWS, Docker, CI/CD*
- Implemented a **containerized REST pipeline** with automated CI/CD validation, increasing production reliability for large-scale transformation workloads.